MACE: Masking-Aware Contract Engine

Current pipeline (M13, 2026-11-04) — operation-level MSMS / MMBS discovery with optional semantic hypotheses

0 | INPUTS

program P, marked operation A, finite legal domain D

candidate operations from AST + graph: U_graph

optional

1a | Deterministic discovery (C0)

AST SourceIndex + graph edges

executable statements -> U_graph

no model call

1b | Optional semantic proposal

Qwen 32B proposes MSMS / MMBS sets

condition + source evidence + hypotheses

status = HYPOTHESIS only

2 | AST identity mapping

U_union = U_graph + U_qwen (U_graph is never pruned)

new executable statement -> U_qwen

if / for / while / try / with / match -> control_context (never deletable)

U_union

3 | Role-separated causal search (real P+ / P- execution)

MSMS (masked side)

queue: kept set S

evaluate D = U - S

minimal nonempty S with Masked(D)

MMBS (exposed side)

queue: disabled set H

evaluate D = H

minimal nonempty D with Exposed(D)

shared EvidenceStore · strict alternation · probe budget · OK / UNKNOWN / INVALID · UNKNOWN is never False

evidence

4 | Bounded verification (sole confirmer)

only_set_masks · disabling_set_exposes · proper-subset minimality

condition exact / sound / complete on the pilot domain

real v3: set-level confirmation; condition NOT_EVALUATED

model output cannot promote itself

not confirmed

4b | Feedback (max 1)

real failed input + arguments

condition truth / set evidence

stripped control / mapping errors

feedback loop (max 1)

VERIFIED / SET_CONFIRMED

HIDDEN reference / scorer

evaluation only

never enters prompts, search, or confirmation

5 | OUTPUT

confirmed MSMS / MMBS per input + conditions

evidence: BOUNDED / TESTED / REFUTED / UNKNOWN — never PROVEN

unconfirmed = UNKNOWN / NOT_EVALUATED, not False

solid blue: deterministic

dashed orange: optional Qwen

green: confirmed output